

Deep Learning-Based Stoma Image Classification - Public Technical Excerpt

Public-safe technical excerpt derived from the original project output.

Source: Original industry-academic project presentation (Ajou University ML & Data Mining Laboratory)

Project focus

Classification of patient-captured stoma images with explicit model-interpretability and robustness analysis.

Baseline result

EfficientNet-B0 baseline AUC: 0.8919 +/- 0.0057. Stratified 5-fold evaluation was reported in the project presentation.

Interpretability finding

Grad-CAM analysis indicated that the classifier could attend to surrounding gauze and acquisition context rather than the clinically relevant stoma region, revealing background bias.

Optimization

Subsequent ROI-centered preprocessing and Label Smoothing improved the portfolio-reported AUC to 0.92. This result is kept separate from the presentation-stage baseline.

Public release scope

No patient-captured clinical image or raw clinical data is redistributed. Selected figures in the repository are public-safe technical reconstructions based on the original analysis.